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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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PROJECT TYPE: **DBMS PROJECT** STUDENT: **NISHANTH KR**

PROJECT GUIDE: **Ms.Abishree.S AP/CSE**

PROJECT TITLE: **GLOBESENSE: REAL-TIME 3D GEOSPATIAL TRACKING AND VISUALIZATION PLATFORM**

Abstract

GlobeSense is a high-performance, browser-based geospatial intelligence and database-driven visualization platform that unifies real-time public telemetry streams onto a single interactive 3D virtual globe. Traditional tracking systems operate within isolated, single-purpose web silos, forcing analysts, students, and operators to toggle between disparate applications to monitor flights, maritime vessels, orbital satellites, seismic events, and atmospheric conditions. GlobeSense replaces these fragmented interfaces with an integrated operational command console, enabling fluid transitions from a planetary macro-overview down to individual aircraft vectors, naval hulls, and municipal street infrastructure.

The client architecture features multi-layered geospatial data filtering, an interactive click-to-track engine that computes camera look-at vectors and renders fading kinematic trajectories, a first-person cockpit simulation over authentic terrain elevation models, and custom post-processing visual sensor modes (Night Vision, Thermal FLIR, and CRT Phosphor). Analytical capabilities include geodetic distance measurement, polygon surface area calculation, turn-by-turn multi-modal routing, and meteorological overlays including dynamic wind particle velocity vectors, precipitation radar, cloud reflectance, lightning density, and tropical cyclone trajectories.

Engineered as a distributed client-server ecosystem, GlobeSense integrates an embedded SQLite relational database for mission parameters, DuckDB Spatial SQL executing analytical queries over columnar Apache Parquet archives, and a secure Node.js proxy server that protects upstream API secrets, sanitizes queries, and prevents client-side performance degradation. The platform ingests telemetry from open public repositories—including OpenSky Network, adsb.lol, CelesTrak, USGS, NASA FIRMS, NOAA, and OpenStreetMap—operating entirely without mandatory paid API keys. Optional free keys activate photorealistic 3D urban tilesets and real-time maritime AIS tracking, demonstrating an accessible and production-grade approach to database-driven geospatial analytics.

Objectives

- To architect a centralized, database-backed web application unifying heterogeneous live public telemetry feeds onto an interactive 3D virtual globe.
- To implement a persistent relational database schema for indexing satellites, flights, ground stations, CCTV nodes, and user annotation layers.
- To integrate spatial SQL query processing via DuckDB and Parquet storage for high-speed spatial predicate evaluation and geospatial boundary filtering.
- To engineer real-time entity tracking with dynamic camera orientation, fading historical trails, and synchronized live telemetry HUD cards.
- To build a realistic first-person cockpit simulation mode synchronizing real-time flight vectors with 3D digital elevation models.

- To develop custom GPU-accelerated GLSL shader pipelines providing multi-spectral visual sensor modes (thermal infrared, night vision, and CRT).
- To integrate real-time atmospheric and meteorological observation layers, including Doppler radar, wind velocity fields, and cyclone forecasting.
- To implement SGP4 orbital propagation algorithms for real-time satellite trajectory calculation based on CelesTrak Two-Line Element (TLE) sets.
- To construct a resilient Node.js middleware proxy implementing request rate-limiting, spatial bounding-box clipping, and credential isolation.
- To deploy the platform on cloud infrastructure with zero mandatory paid subscription requirements, ensuring open educational and research accessibility.

Work Completed

This project is built upon the open-source God's Eye View architecture (MIT License), significantly extended and customized with robust spatial database pipelines, telemetry ingestion layers, and custom client interfaces. The core work completed includes:

- Database Schema & Relational Telemetry Storage:
  - Designed and deployed the SQLite relational database schema (satellite-operations.sqlite) defining structured tables for satellites, flights, cameras, crew, and data_layers.
  - Implemented primary key constraints, foreign key referential integrity, and timestamped audit logs for mission telemetry and tracking sessions.
  - Configured DuckDB Spatial queries over partitioned Apache Parquet datasets to execute spatial filtering (ST_PointOnSurface, ST_Area_Spheroid) and bounding box clipping.
  - Integrated Overpass QL relational querying over OpenStreetMap datasets to extract regional infrastructure, runways, and military coordinates.
- Multi-Stream Live Telemetry Ingestion Pipeline:
  - Implemented automated ingestion handlers for ADS-B flight data (OpenSky Network & adsb.lol REST APIs) with state deduplication.
  - Integrated CelesTrak TLE orbital ephemeris feeds utilizing SGP4 orbital propagation algorithms to calculate real-time satellite positions.
  - Integrated USGS real-time seismic event feeds and NASA FIRMS active thermal wildfire hotspot detection data.
  - Connected live maritime AIS vessel tracking, public CCTV camera feeds, transit networks, and micro-mobility bikeshare data.
- 3D Geospatial Globe & Coordinate System Engine:
  - Implemented high-performance 3D rendering using CesiumJS, integrating WGS84 ellipsoidal geometry and EGM96 geoid gravitational height corrections.
  - Configured multi-source basemap switching across Esri World Imagery, OpenStreetMap vector tiles, and Google Photorealistic 3D Tiles via Cesium ion.
  - Developed custom GLSL post-processing shaders delivering immersive sensor vision modes: Night Vision (green phosphor), Thermal FLIR (false-color heat map), and CRT Tactical Scanlines.
- Interactive Tracking, Cockpit Simulation & Telemetry Cards:
  - Built click-to-track camera locking with orientation calculation, heading interpolation, and fading polyline flight trails.
  - Developed first-person cockpit simulation mode projecting real elevation terrain beneath moving aircraft models.
  - Created real-time telemetry HUD overlay displaying callsign, aircraft type, altitude, ground speed, heading, and geographic coordinates.

- Drawing Tools, Spatial Routing & Mission Recording:
  - Implemented geodetic measurement tools for real-time calculation of surface distances, bearings, and enclosed polygon areas.
  - Integrated OpenStreetMap Nominatim and OSRM routing engine supporting turn-by-turn driving, walking, and cycling route navigation.
  - Developed Scene Director timeline playback engine enabling users to record, script, and replay cinematic geospatial flythrough tours.
- Security, Proxy Middleware & Deployment Architecture:
  - Engineered a resilient Node.js proxy server enforcing client rate-limiting, error sanitization, and upstream API credential encapsulation.
  - Configured automated CORS policy handling and response caching to minimize upstream API quotas and latency.
  - Deployed production client and proxy server on Vercel with high-availability edge routing.

Current Status

The GlobeSense platform is fully functional and live in production on Vercel. It seamlessly renders 19 interactive geospatial data layers across the globe, executing real-time entity tracking, live telemetry HUD updates, cockpit perspective simulation, GLSL multi-spectral visual filters, weather intelligence displays, spatial vector annotation tools, and turn-by-turn route calculations. The application operates without requiring any paid API subscriptions, with free optional API keys unlocking photorealistic Google 3D Tiles and maritime vessel tracking.

On the database tier, the embedded SQLite database engine (satellite-operations.sqlite) manages relational records for satellites, active flights, traffic cameras, flight crews, and layer configurations with full referential integrity. DuckDB Spatial executes analytical SQL queries against Apache Parquet vector archives, while the Node.js middleware proxy manages upstream connection pooling, response payload caching, and secure API credential isolation. The platform maintains sub-second query execution and a fluid 60 FPS rendering rate on modern web browsers.

Technologies Used

Frontend 3D Engine: CesiumJS (v1.124), WebGL, GLSL Custom Shaders

Web Technologies: JavaScript (ES6+), HTML5, CSS3, Vite 6.0 Bundler

Database Engines: SQLite 3 (satellite-operations.sqlite), DuckDB Spatial (Parquet SQL)

Backend & Middleware: Node.js (Proxy Server), RESTful JSON Endpoints, WebSockets

Geospatial & Math Libraries: Satellite.js (SGP4), EGM96-Universal, MGRS, Mapbox Vector Tile

External Data Feeds: OpenSky Network, adsb.lol, CelesTrak, USGS, NASA FIRMS, NOAA, OpenStreetMap

Map Providers & 3D: Esri World Imagery, OpenStreetMap, Google Photorealistic 3D Tiles (Cesium ion)

Hosting & Deployment: Vercel (Production Edge Network Hosting)

Technology Justification:

CesiumJS delivers an authentic WGS84 ellipsoidal globe with true geodetic coordinates, dynamic elevation clamping, and atmospheric physics that 2D mapping libraries cannot represent. The combination of SQLite and DuckDB Spatial enables zero-configuration, high-performance relational and columnar spatial SQL processing directly on the server without incurring the substantial financial and operational overhead of dedicated cloud database instances. Node.js provides non-blocking asynchronous I/O ideal for proxying high-frequency telemetry streams, while Vite ensures rapid hot-module replacement and optimized client-side bundle delivery.

Project Plan for Future

- Implement historical telemetry replay using time-partitioned SQLite tables to visualize historical flight paths and naval routes.
- Integrate PostGIS / PostgreSQL cloud database replication for enterprise-grade collaborative multi-user mission tracking.
- Add real-time spatial geofencing alerts triggered via automated spatial queries when tracked entities enter designated airspace.
- Incorporate ADS-B Mode-S military aircraft transponder filters and automated anomalous squawk code (7500, 7600, 7700) detection.

- Expand meteorological modeling by integrating high-resolution GRIB2 weather forecast datasets via WebAssembly decoders.
- Integrate hands-free speech recognition and AI voice command control for camera navigation and layer toggling.
- Develop automated PDF and CSV reporting modules for generating situational intelligence briefings and mission logs.
- Optimize mobile browser touch gestures and responsive HUD layout for mobile and tablet field operation.

Expected Outcome

The expected outcome of the GlobeSense project is a production-grade, highly dependable 3D geospatial intelligence platform that bridges disparate public data streams into a unified situational awareness console. By replacing fragmented tracking tools with an integrated virtual globe, users gain immediate access to cross-domain activity encompassing aviation, maritime, orbital, seismic, and meteorological environments.

From a DBMS perspective, the project demonstrates how modern lightweight database architectures—combining SQLite relational integrity with DuckDB spatial analytics—can process high-velocity geospatial datasets efficiently without prohibitive infrastructure costs. Administrators and researchers benefit from sub-second coordinate resolution, spatial boundary queries, and secure credential handling via the proxy layer.

The platform delivers significant educational, analytical, and practical value, empowering students and researchers to study orbital mechanics, air traffic flow, maritime logistics, and disaster monitoring within an intuitive, open-access browser environment. By operating entirely on free and open tiers, GlobeSense establishes a sustainable and replicable blueprint for modern geospatial systems.

Conclusion

GlobeSense demonstrates the successful fusion of relational database management, spatial SQL analytics, and cutting-edge 3D WebGL graphics into an interactive planetary digital twin. By bringing together flights, ships, satellites, earthquakes, wildfires, and weather phenomena onto a single globe, the system proves that open public data feeds can deliver rich, institutional-grade intelligence when paired with an optimized backend.

The utilization of CesiumJS, SQLite, DuckDB Spatial, and a secure Node.js proxy establishes a modular, secure, and maintainable software architecture. GlobeSense exemplifies how contemporary database principles and modern web development can automate complex geospatial tracking workflows, providing an exceptional and immersive user experience.

STUDENT **PROJECT GUIDE** **HOD**